

CS4800: Designing Sustainable ICT Systems

Lecture 2: Defining Sustainability

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What will you learn in this lecture?

Lecture 2: Defining Sustainability

- Conceptualizing sustainability
- Defining sustainable ICT systems
- Frameworks for sustainability: planetary boundaries, circular economy
- Sustainability indicators: global warming potential, emissions reporting, carbon/energy/material/water intensity

Sustainability: history and context

- From Latin *sustinēo* - to uphold, support, maintain, or preserve
- [*Silvicultura oeconomica*](#) (1713): long-term responsible use of forest resources
- [*Normative*](#) concept: not how the world is, but how it should be

DESCRIPTIVE VS. NORMATIVE

- **Descriptive** claims concern *what is, was, or could be*
 - Scientific inquiry, for example, seeks to accurately describe and predict phenomena that exist in the world or universe.
 - Example: *How* do we genetically modify animals?
- **Normative** claims concern *what should be*
 - Facts matter to ethics, but the aim of ethical analysis is to figure out what should be the case (even if it does not actually turn out that way).
 - Example: *Should* we genetically modify animals? What are the *moral* costs and *moral* benefits of the available options?



[Descriptive vs normative claims](#)

Definitions: sustainable development

- [Brundtland Commission](#) (1987): meet the needs of the present without compromising the ability of future generations to meet their own needs
- [Natural Step](#) (1989): limit extraction and artificial substances, preserve environment while meeting human needs



[Gro Brundtland, Norwegian prime minister and chair of the Brundtland Commission](#)

Definitions: planetary spheres, impacts

1. Atmosphere: [greenhouse gas emissions](#)
2. Hydrosphere: [over-extraction](#), [nitrogenation](#)
3. Pedosphere: [industrial agriculture](#)
4. Lithosphere: [over-mining](#)
5. Biosphere: [habitat loss](#)
6. Anthroposphere: [debt burden](#)



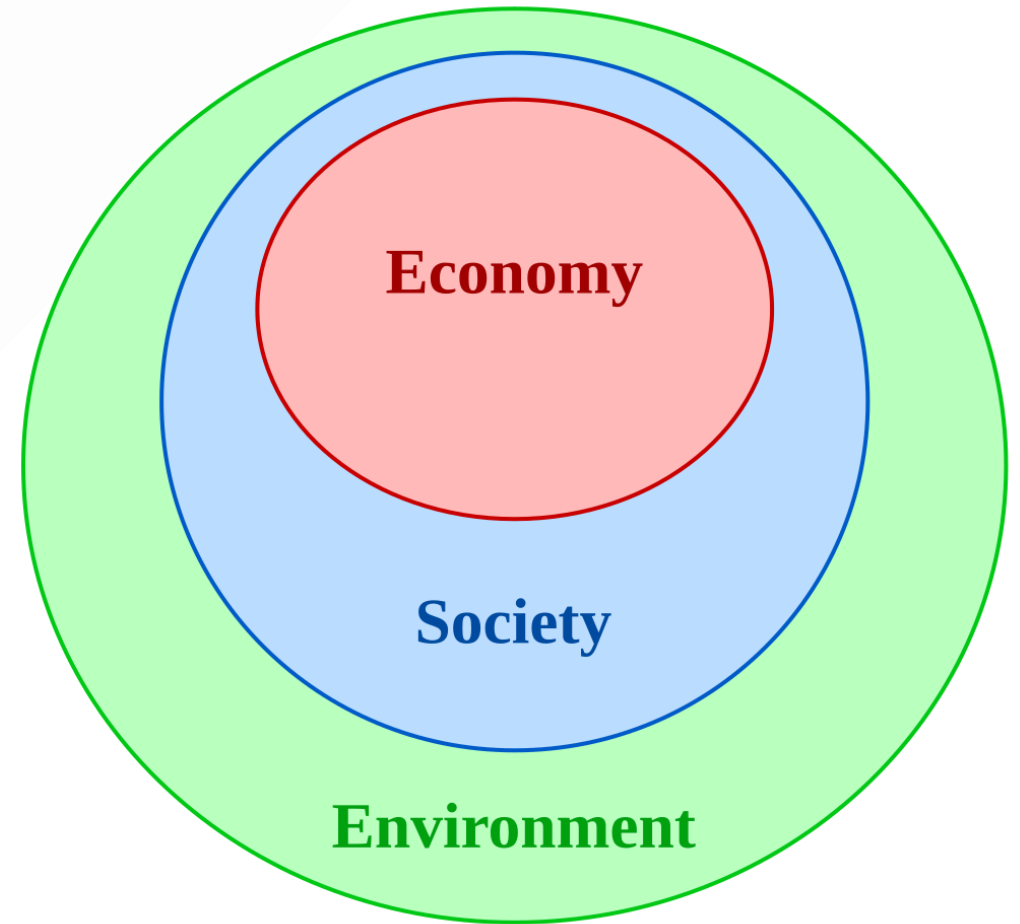
[Four recognized planetary spheres](#)

Definitions: dimensions of sustainability

[Rio Declaration](#) (1992): three [dimensions](#)
(environmental, social, economic)

1. **Environmental:** stewardship of natural resources for future generations
2. **Social:** ability of society to continue existing without [structural obstacles](#)
3. **Economic:** policies that support long-term [viability](#) without present-day [damage](#)

Hierarchy: environmental sustainability [overarching](#) → we will (mostly) focus on it when we talk about sustainability



[Hierarchy of the dimensions of sustainability](#)

Defining sustainable ICT systems

Aspects shared by previous definitions:

1. Sustainability is an objective to be reached
2. Sustainability involves the protection of the planetary system
3. Sustainability involves judicious use of resources

-> **Sustainable ICT systems minimize impact on natural resources while meeting societal needs**



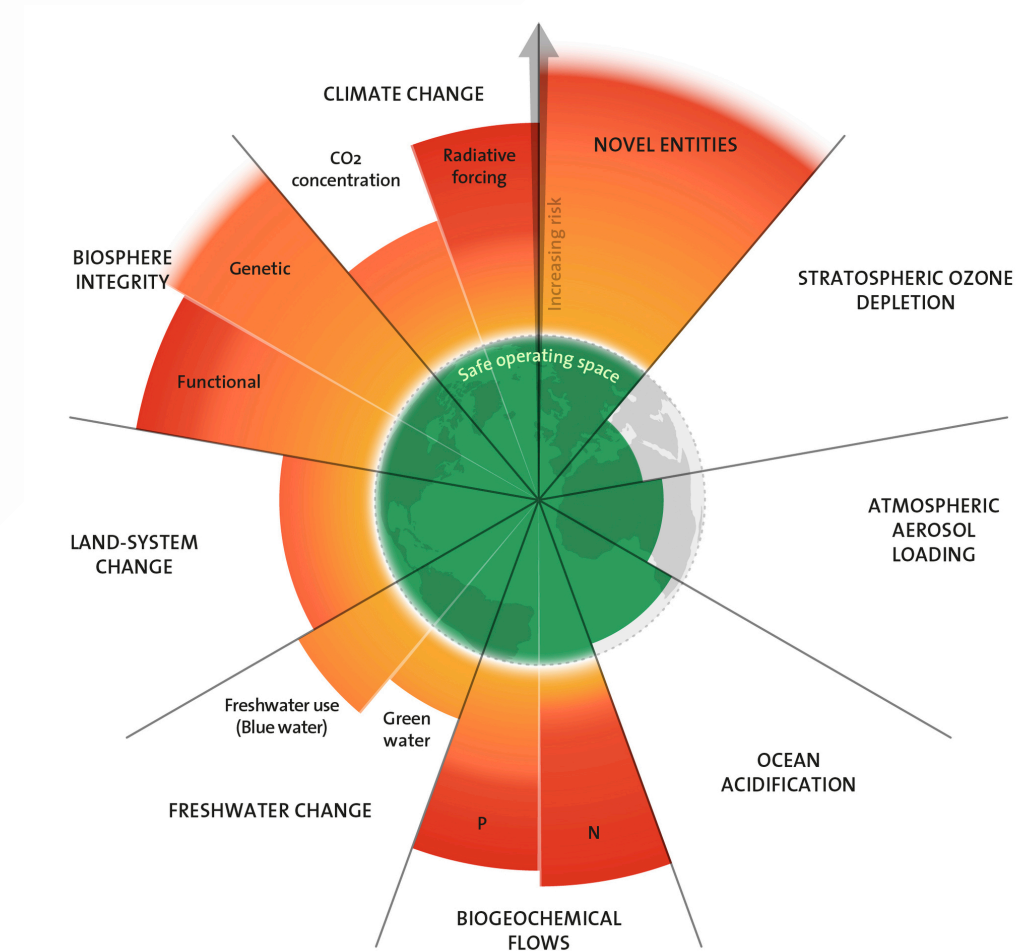
Hacker voice: I'm in

What frameworks can we use to think about sustainability?

Planetary boundaries

[Stockholm Resilience Centre](#) (2009): safe limits for human pressure on critical processes

1. Climate change: [429.64](#) ppm CO₂
2. Biosphere integrity
3. Ocean acidification
4. Nitrogen, phosphorus flows
5. Freshwater change
6. Land system change
7. Stratospheric ozone depletion
8. Atmospheric aerosol loading
9. Novel entities: e.g. microplastics



[Quantifying human pressures on each planetary boundary](#)

Precautionary principle

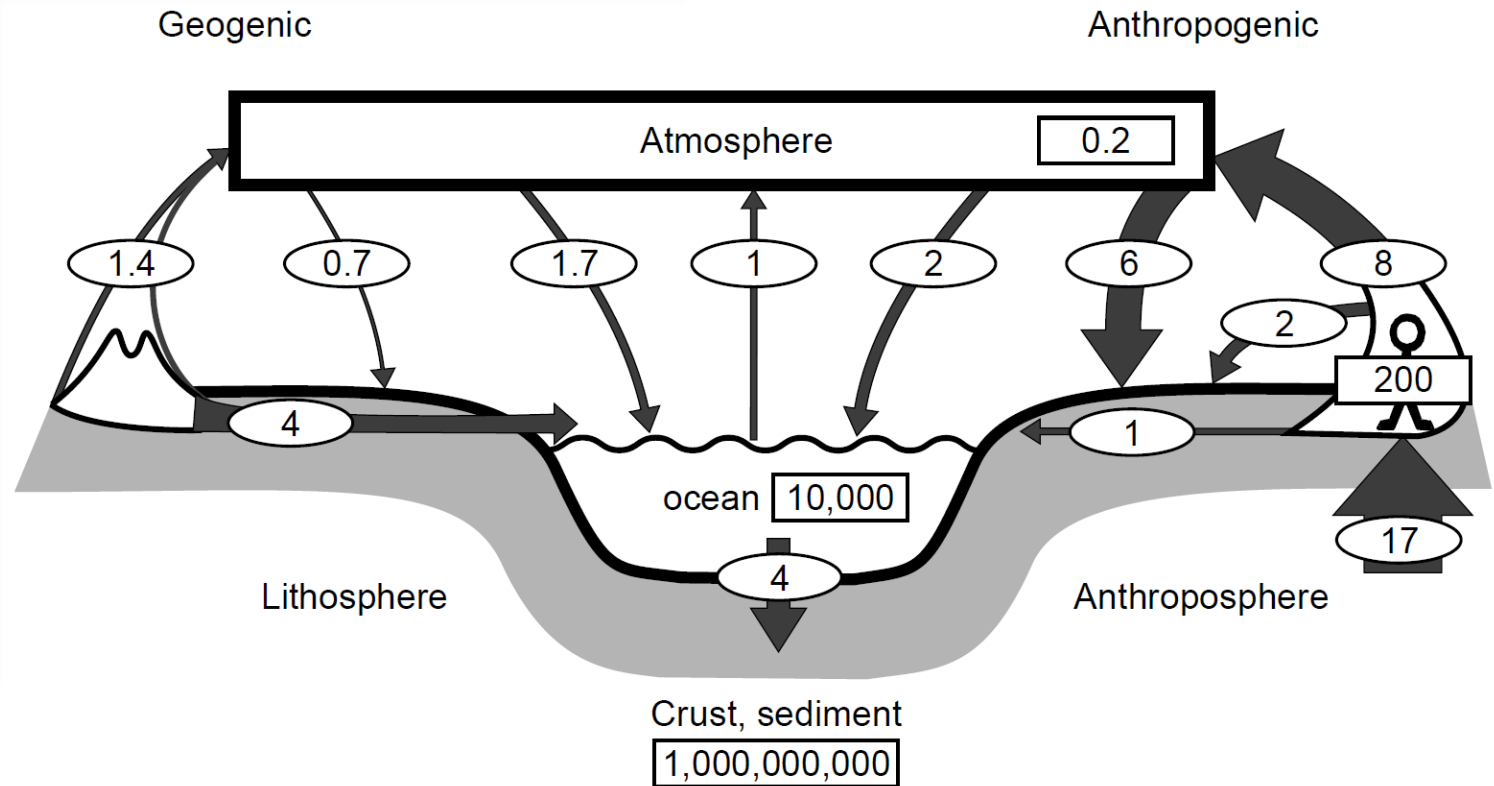
- Complex planetary system → great [uncertainties](#) in calculating exact limits
- [Rio Declaration](#) (1992): if we risk serious damage, insufficient certainty is not a reason to delay preventive measures



[Burden of proof vs precautionary principle reasoning](#)

Tools: material flow analysis

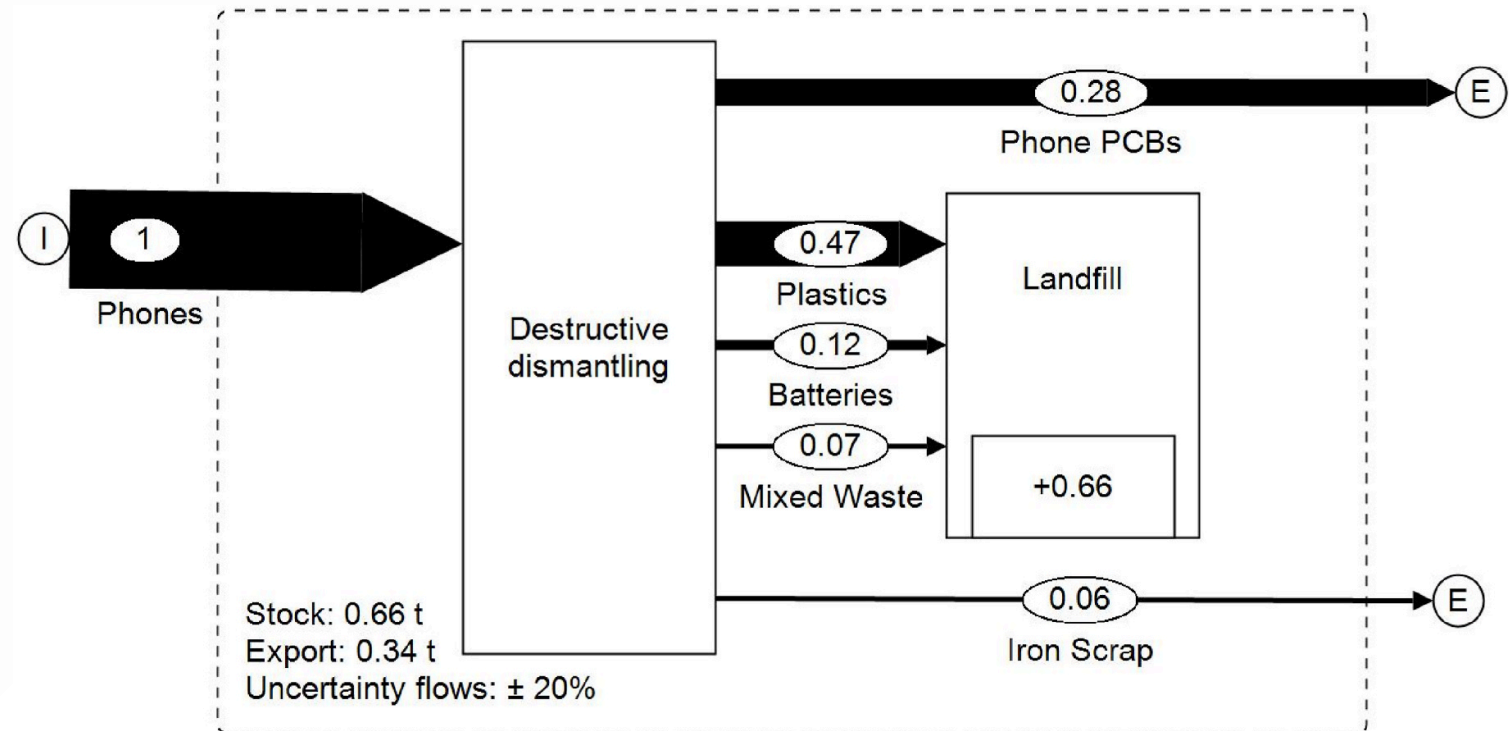
- Developed to measure physical impact of human society on natural systems
- Calculates how individual substances (e.g. carbon, copper) or materials (steel, plastics) travel through economic and natural system
- [Focus on](#) stocks (static) and flows (dynamic)



[Global cadmium stocks and flows in the 1980's in kilotons - note human-made flows on the right](#)

Example: phone recycling

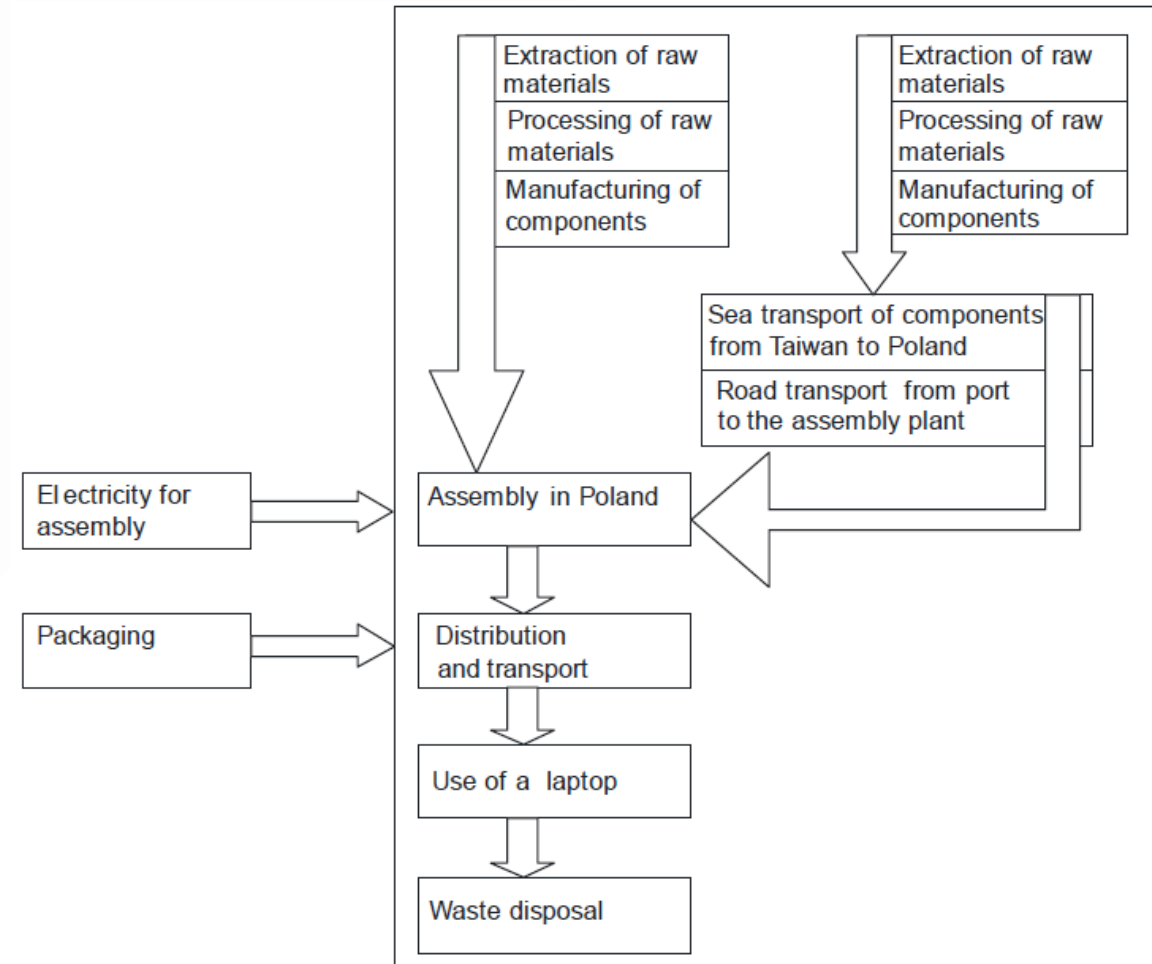
- What happens when a kilogram worth of phones is dismantled?
- Largest part of components (case, battery etc) go to landfill
- Some components (PCB, structure) further processed
- MFA gives us information about where raw materials in ICT devices end up



Material flows and stocks caused by recycling of phones

Tools: lifecycle analysis

- [Environmental footprint](#): total human pressure on [natural environment](#)
- Individual product footprint: calculated through [lifecycle analysis](#)
- Environmental impact calculated for every activity: materials sourcing, manufacturing, distribution, decommissioning etc



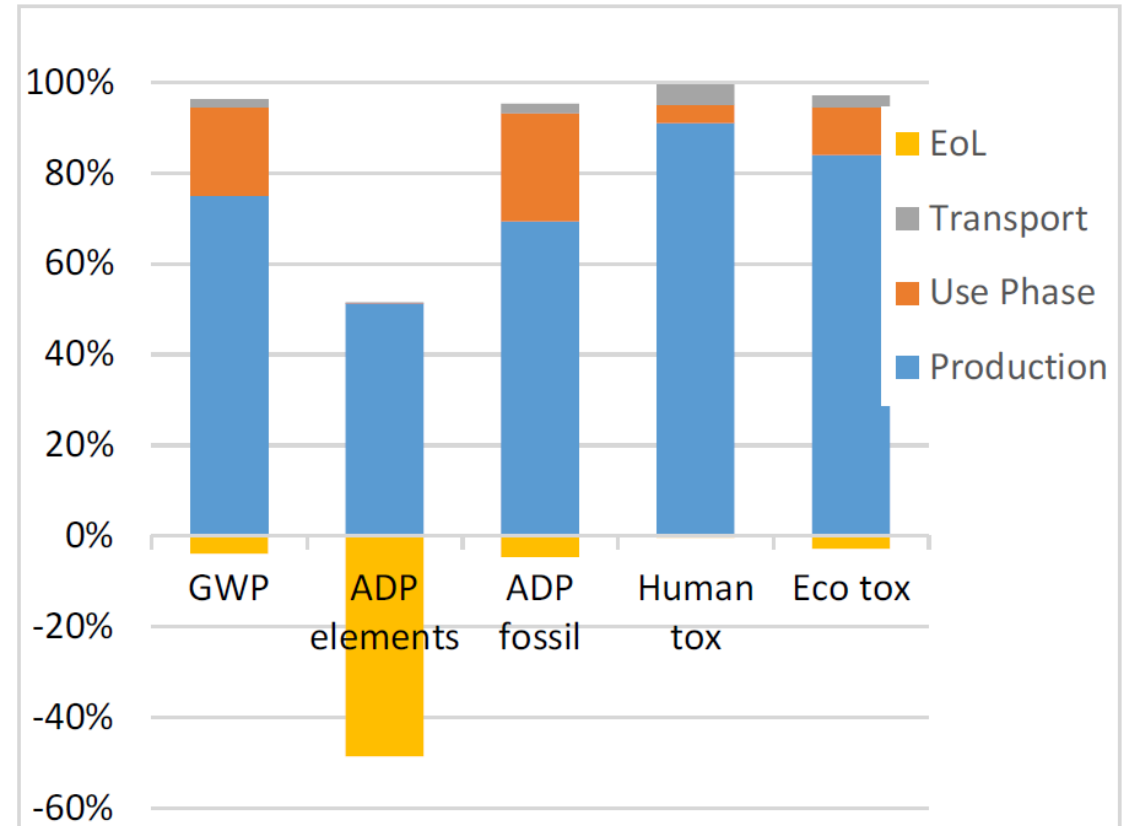
[Example: processes involved in the lifecycle of a laptop](#)

Example: Fairphone LCA

External contractor [quantified](#) product's environmental footprint over impact categories:

1. Climate change (GWP)
2. Material use (ADP elements)
3. Fossil fuel use (ADP fossil)
4. Human toxicity (Human tox)
5. Environmental toxicity (Eco tox)

Highest impact observed during production phase, distributed across components



[Relative impact \(as percent of total\) for each lifecycle phase on various impact categories](#)

Sustainability indicators

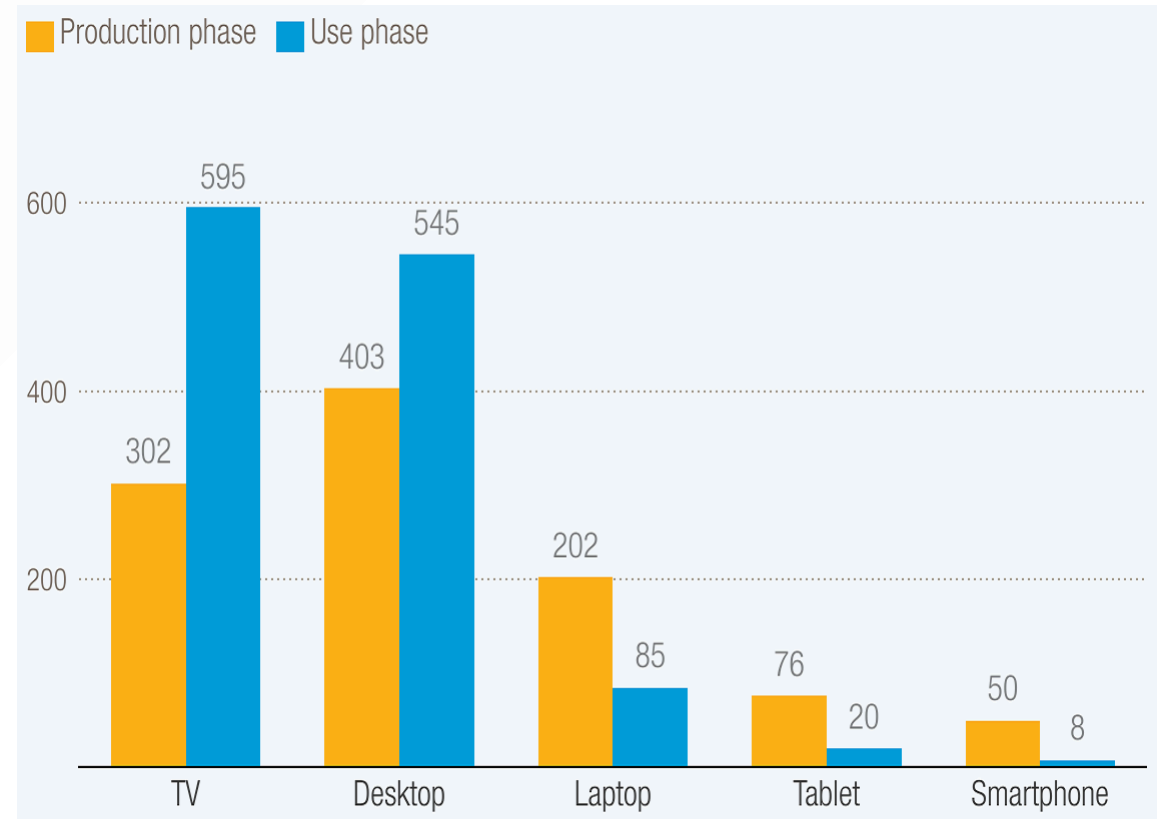
- No universal standard for studying sustainability
- Competing indicators developed
- Indicator: quantitative/qualitative measure of how close we are to a target



Various economic indicators

Sustainability indicators: global warming potential

- [Global warming potential](#): multiple of warming caused by same mass of CO₂ (GWP=1)
- Use GWP to calculate mass of CO₂ that would warm the planet as much as the mass of studied gas: [CO₂-eq](#).
- Use CO₂-eq to calculate carbon intensity: impact of a unit of product or service on climate change



[Average lifecycle emissions of electronic devices in kg CO₂-eq](#)

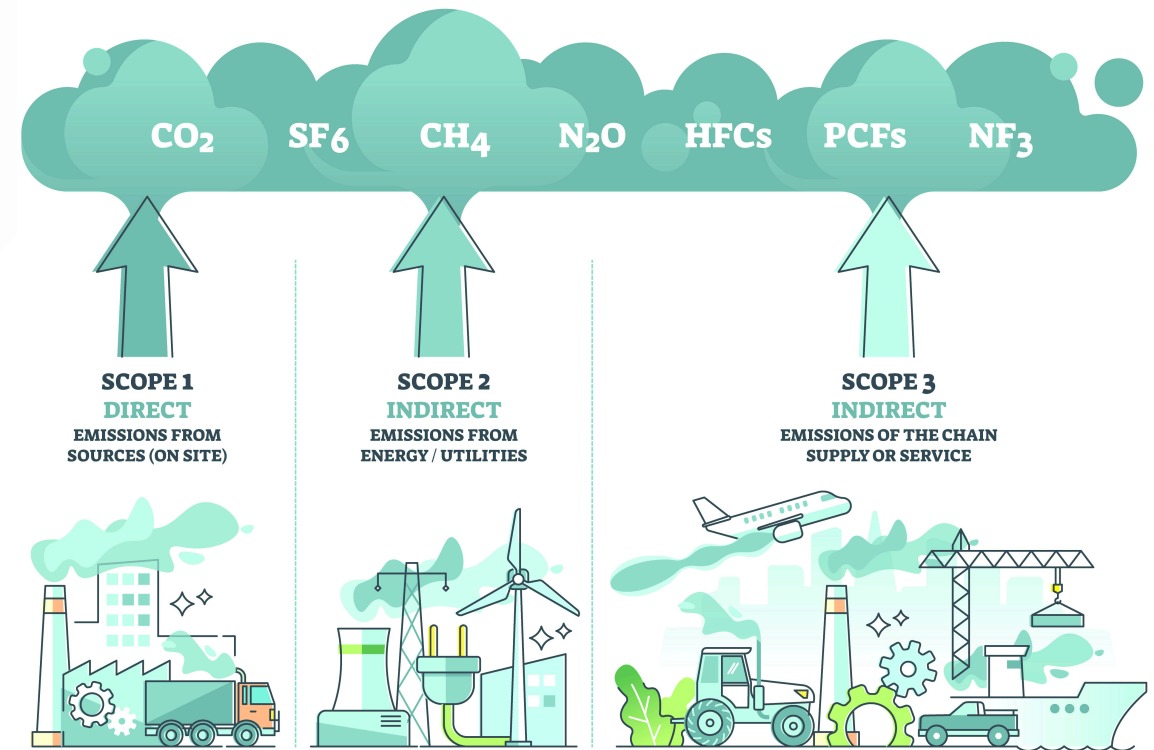
Carbon accounting

- [Tracks](#) an organization's carbon emissions
- National ([US](#), [NL](#)) and international ([EU ETS](#)) monitoring and reporting
- [GHG Protocol](#): define organizational boundary, then [apply](#) scopes:

Scope 1: sources owned/controlled by company

Scope 2: energy consumed by company

Scope 3: indirect emissions, upstream (e.g. manufacturing of goods purchased by company) or downstream (e.g. decommissioning goods sold by company)



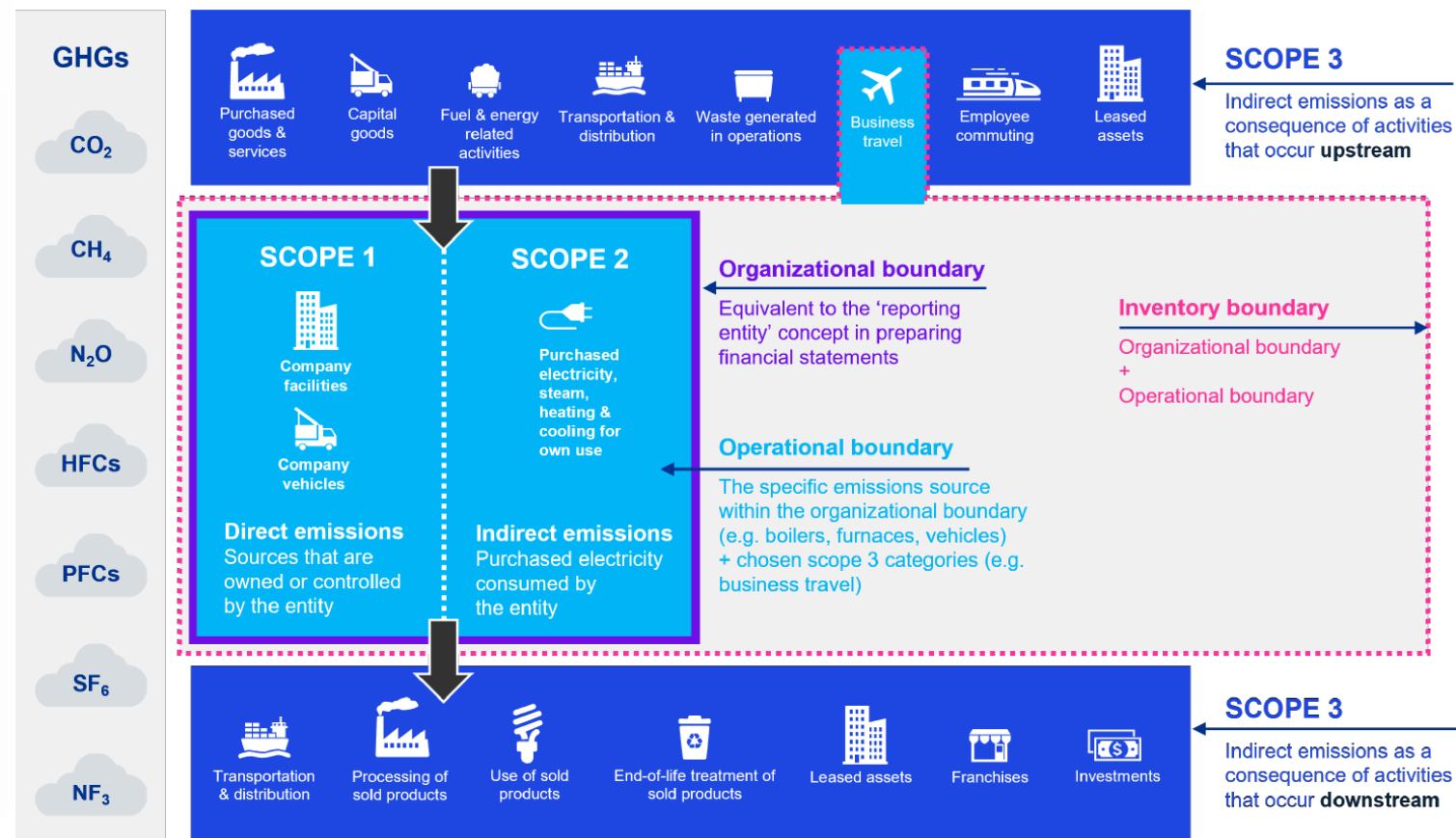
[Scope 1, 2, and 3 emissions](#)

Carbon accounting: GHG inventory

- Total emissions for each scope: multiply activities by carbon intensity:

$$\text{CO}_{2\text{eq}} = \text{Activity} \cdot \text{Carbon intensity} \cdot \text{GWP}$$

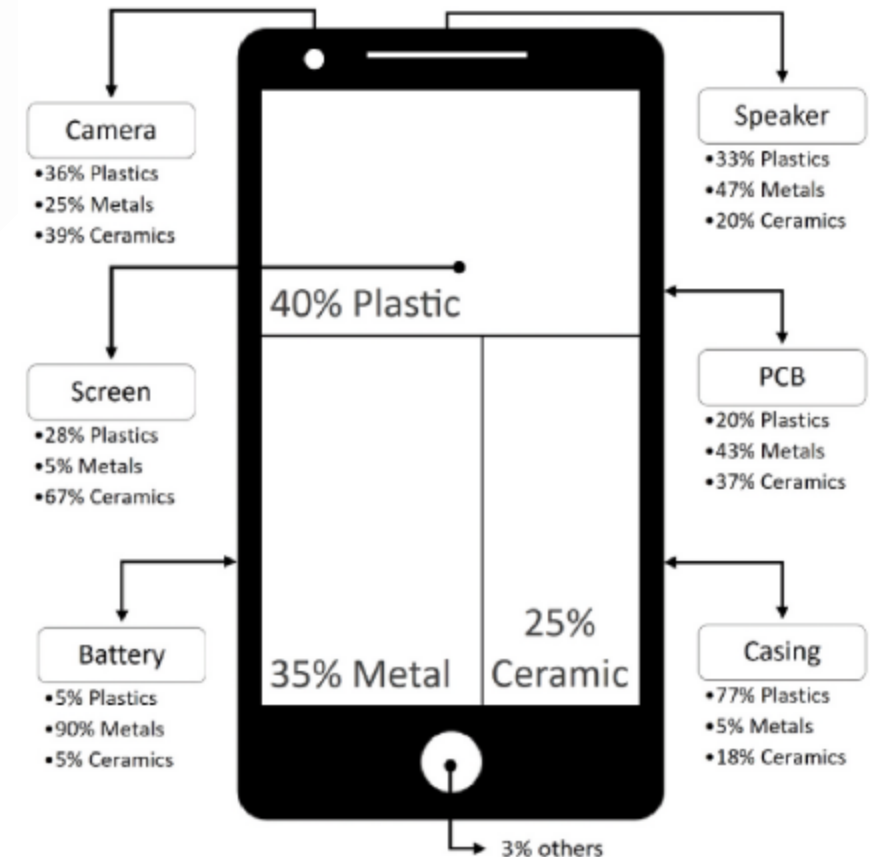
- Emissions data placed in company GHG inventory
- Calculations not always precise: data uncertainty, standard carbon intensity factors, (rarely) malicious underreporting.



Organizational boundary and emissions scopes

Sustainability indicators: material intensity

- [EU 3rd Raw Materials Scoreboard](#): total historical global material use (biomass, fossil fuels, minerals) has exceeded 90 billion tons
- [Critical raw materials](#) (CRMs): supply risk, economic importance
- Significant increase in material use for ICT [expected in the future](#): Pt, Nd, Co etc.
- We use [material intensity](#) (e.g. grams of cobalt per kilogram of phone) to quantify CRM use



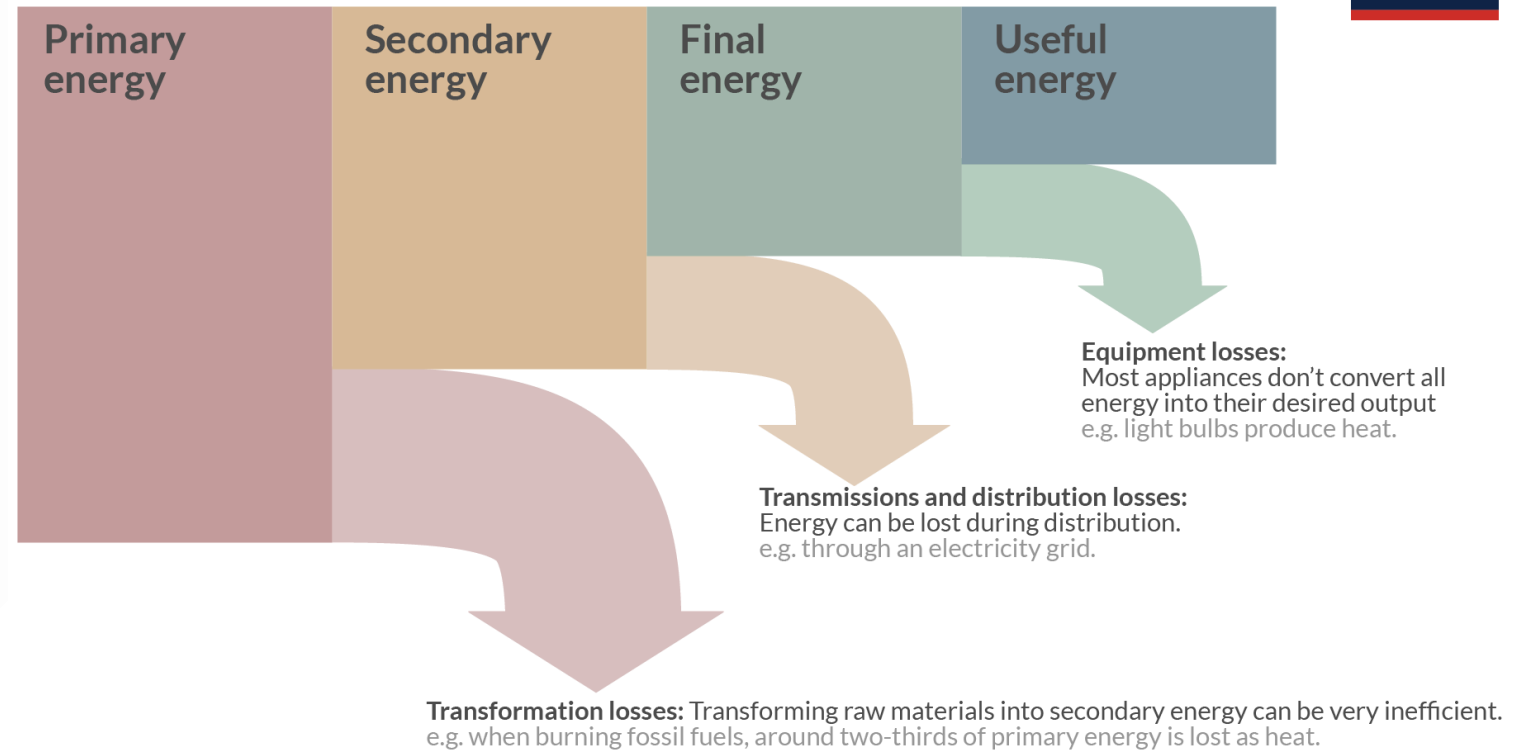
[Material composition of a mobile phone](#)

Sustainability indicators: energy intensity

1. **Primary energy:** found in nature, e.g. chemical energy in coal or oil
2. **Secondary energy:** transportable, e.g. electricity, liquid fuels
3. **Final energy:** used by consumer
4. **Useful energy:** actually used for desired output, e.g. light from lamp

-> (Primary) energy intensity:
energy required per unit of
product/service

Useful energy is just a fraction of primary energy



Our World
in Data

Differentiating primary, secondary, final, and useful energy

Sustainability indicators: water intensity

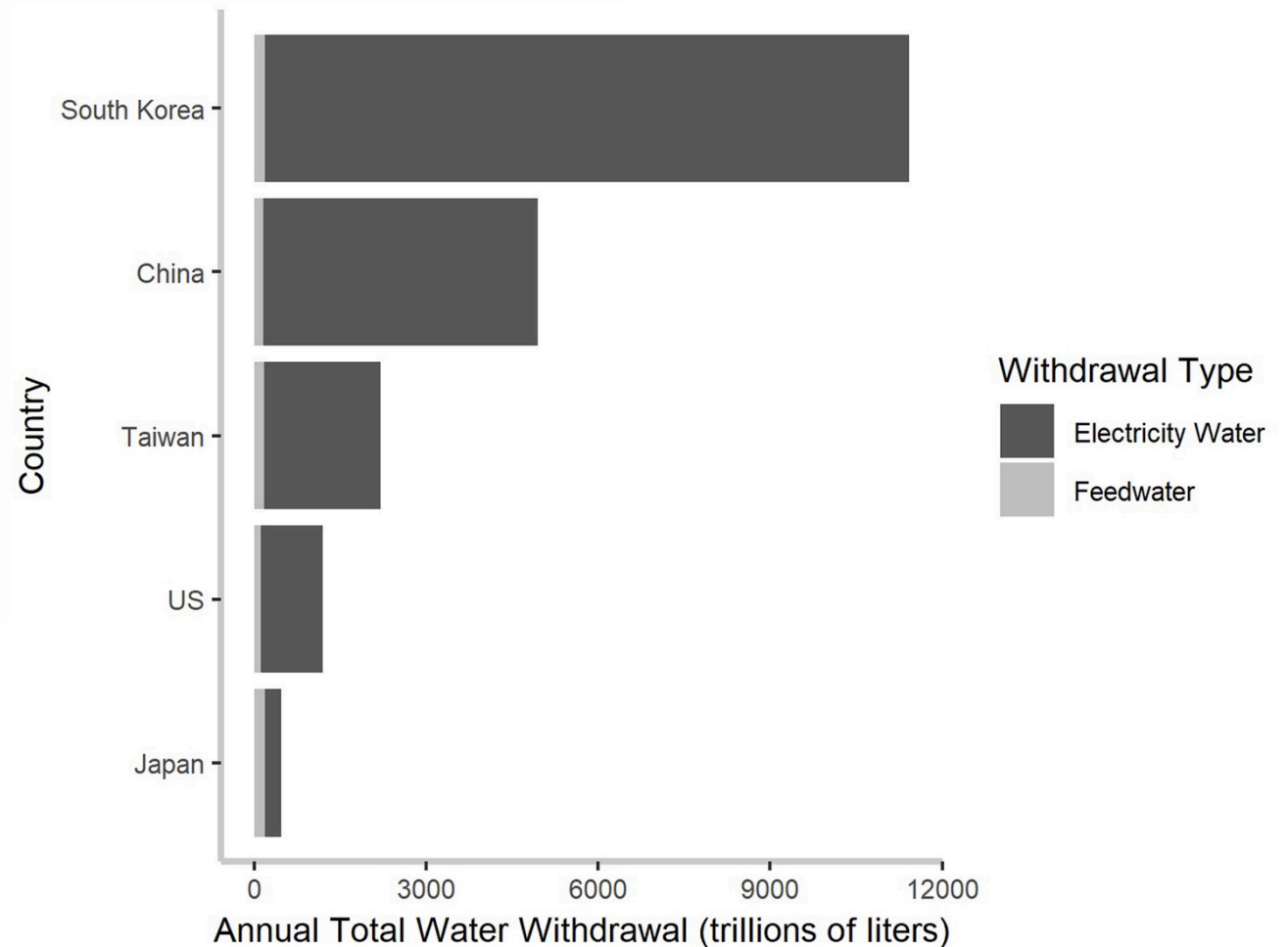
- All life needs water to exist
- Various [indicators](#) used to calculate impact of human system on hydrosphere
- Water intensity: water consumed per unit of product/service
- Not all water used by a process is consumed! Some can be returned as grey water

	125 litres 1 apple		80 litres 1 orange
	50 litres 1 tomato		1260 litres 1 pizza Margherita
	160 litres 1 slice		255 litres 1 glass of milk
	318 litres 100 g cheese		196 litres 1 egg
	2400 litres 1 beef hamburger		132 litres 1 cup of coffee

[Water footprint of various food products](#)

Example: water footprint of semiconductors

- Difficult to calculate water intensity in ICT sector, limited data available
- Semiconductor fabrication for integrated circuits: [most water used for electricity generation](#)
- Results: 7.8 liters/cm² wafer (logic board), 21 liters/cm² wafer (RAM)



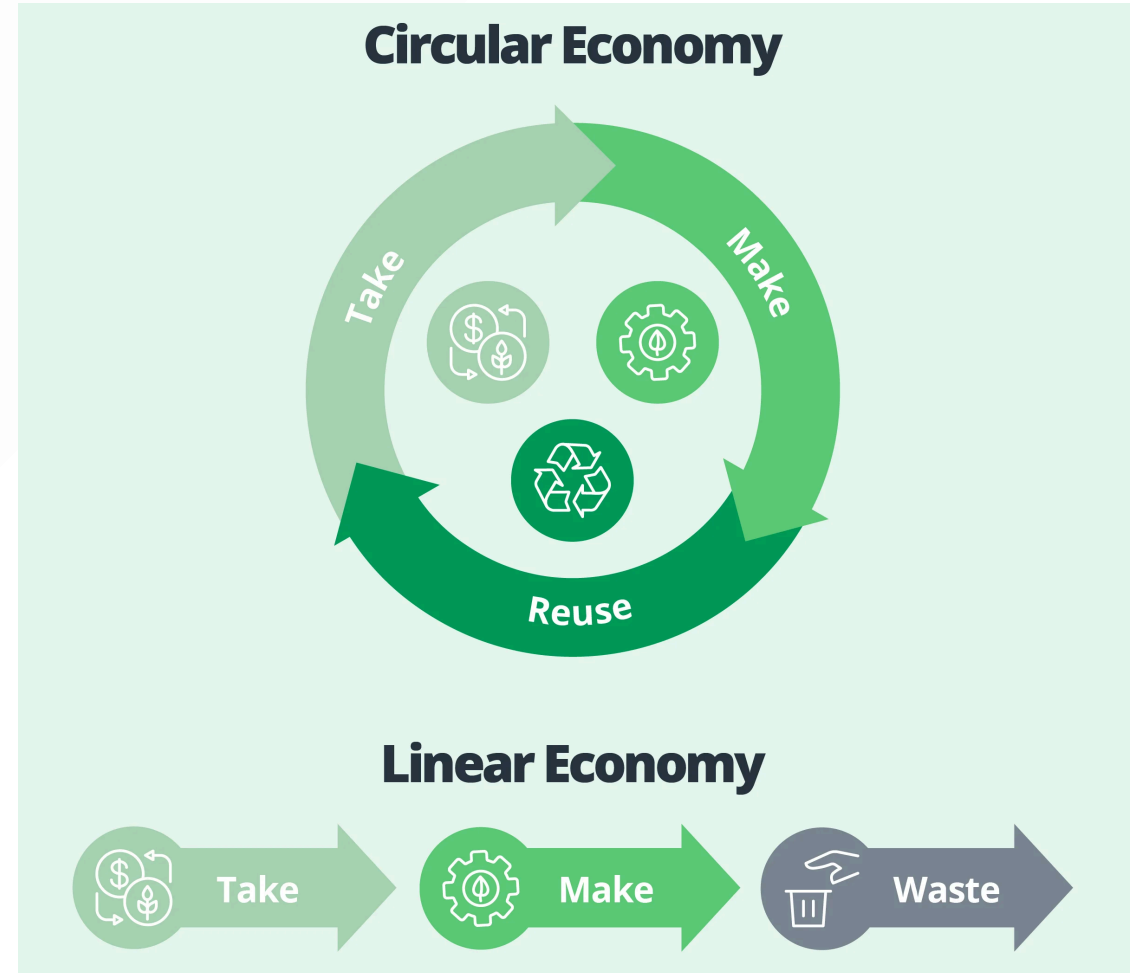
[Water withdrawals for semiconductor fabrication](#)

Sustainability indicators: others

- Various others indicators developed to quantify human interventions in natural systems:
e.g. [biodiversity](#), [ocean acidification](#)
- Not directly relevant to ICT systems

Circular economy: definitions

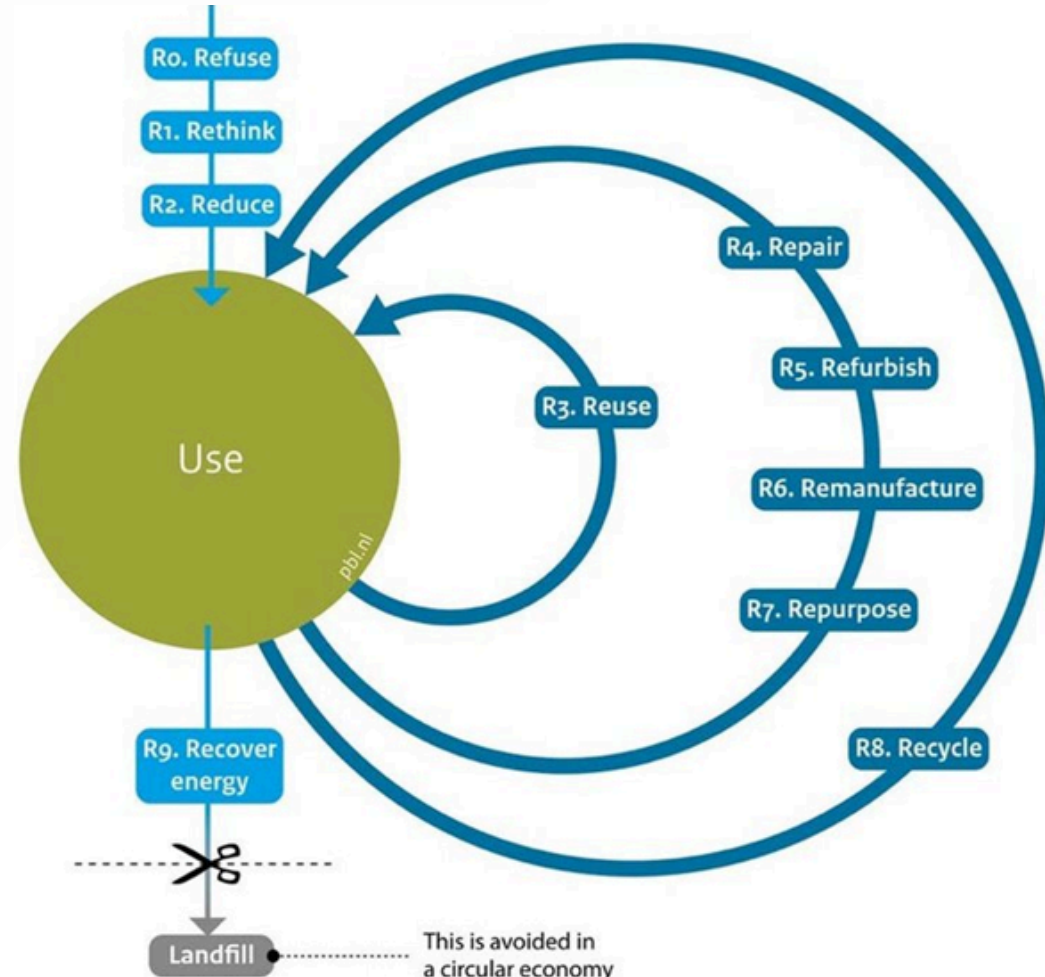
- No universally-accepted definition
- [Ellen MacArthur Foundation](#): economic system with closed loop products/services, allows for long life, reuse, refurbishment, repair, recycling
- [China](#): generic term for reducing, reusing and recycling activities conducted in process of production, circulation and consumption
- Contrast with [linear economy](#): resources processed into products, eventually waste ("take, make, waste")



[Circular vs linear economy](#)

Circular economy: principles

- [Ellen MacArthur foundation](#): objective is to close loops within the economy
- Nine R's: refuse, rethink, reduce, reuse, repair, refurbish, remanufacture, repurpose, recycle, and recover
- Deliberate, reflective design applied to products
- Produce fewer products? What about manufacturing jobs?
- Still aims for economic growth! What about energy coupling?



[The R-ladder used to close loops in the circular economy](#)

Circular economy: decoupling, rebound effects

- Reducing material, energy intensity of products → cost goes down
- Cost goes down → more products can be purchased with same income
- More products purchased → total impacts may actually increase! (Lecture 1)

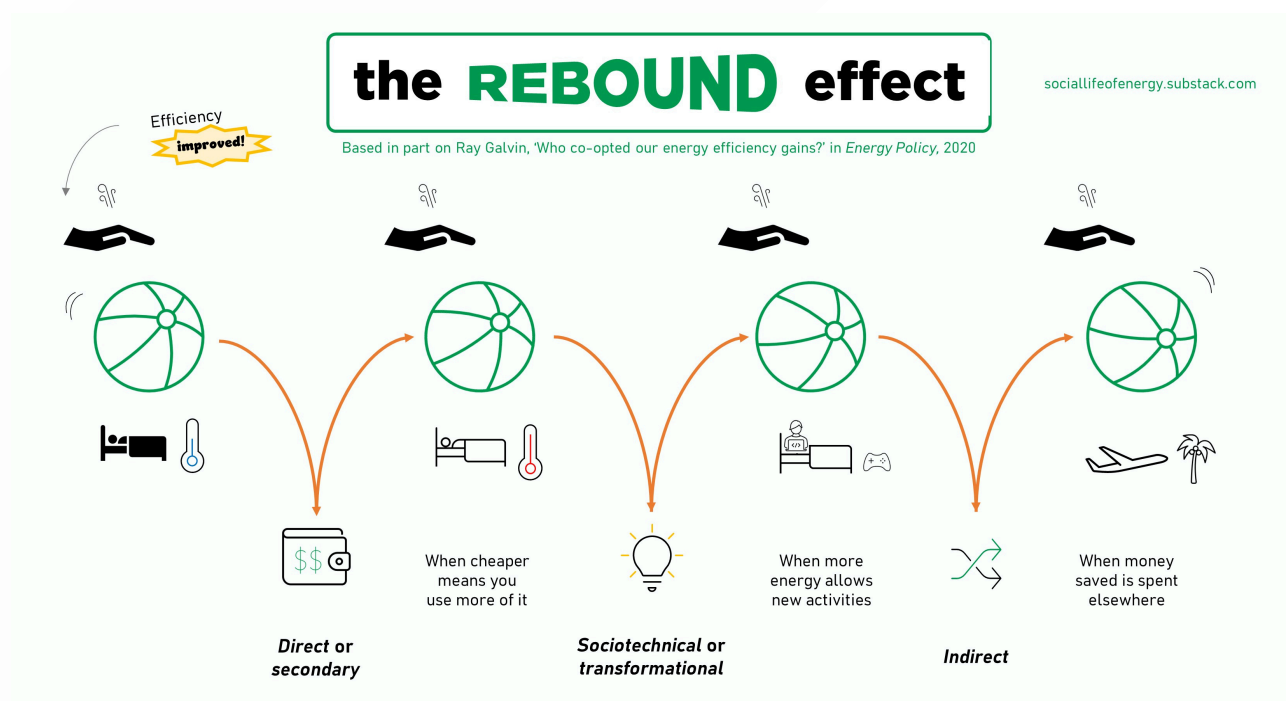


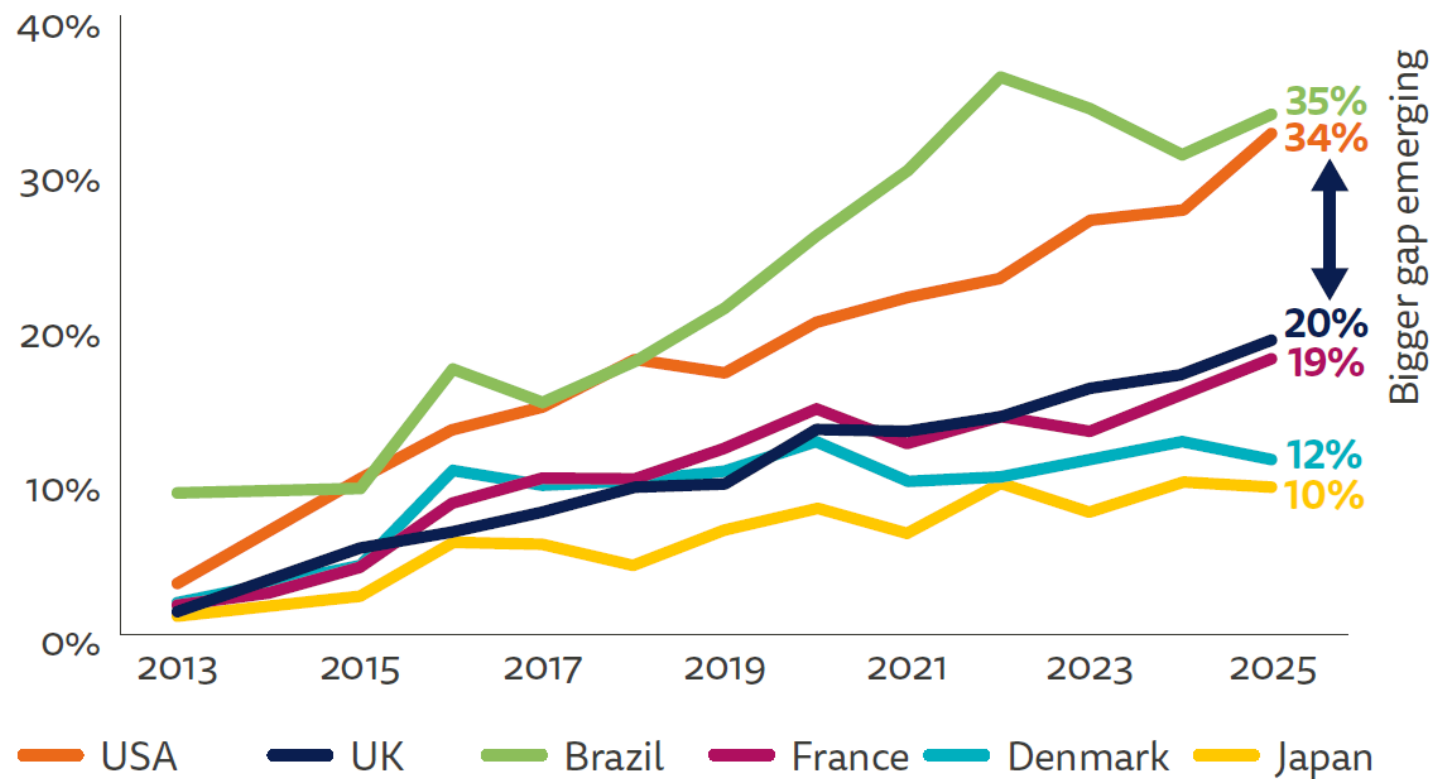
Illustration of the rebound effect

What about social sustainability?

How do we use ICT systems?

- Administration, communication, and entertainment
- [Main gateway to information](#) for majority of people
- [Automation of labor](#), productivity optimization
- Personal, commercial, and public data storage

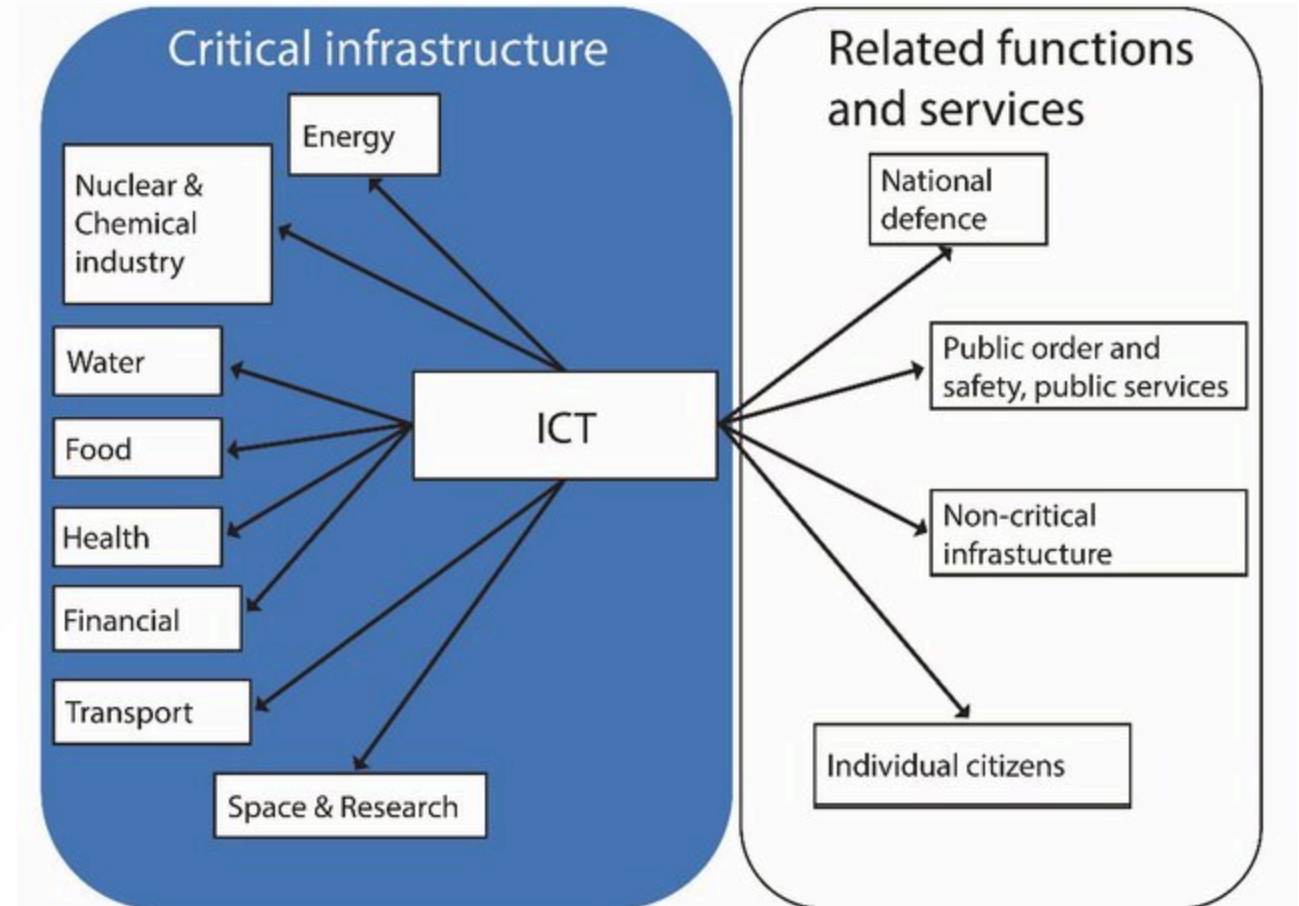
PROPORTION THAT SAY SOCIAL MEDIA IS THEIR MAIN SOURCE OF NEWS (2013-2025) - SELECTED COUNTRIES



[A growing proportion of people get news from social media](#)

Social impacts of ICT

- ICT systems control electricity grid, public transport, administration → critical infrastructure
- Democratization of information has [positive and negative effects](#)
- Centralized data storage empowers [owners over users](#)
- Increasingly digitalized [large-scale surveillance and militarization](#)
- Labor automation can empower workers or [make them redundant](#)



[Social functions met with the help of ICT systems](#)

The human cost of ICT

- How we design ICT systems (energy and material efficiency, repairability) is important
- But also what (and who) we design them for:



Sustainability frameworks and ICT systems

We apply these principles in the lectures to follow:

1. Estimating energy intensity of software (including AI), designing efficient software
2. Evaluating data center energy/carbon/water footprint
3. Studying material intensity of electronics, integrating sustainability in design

Sustainability frameworks allow us to quantify both ICT impacts and mitigation efforts!

Recap

- Sustainable ICT systems minimize environmental impacts while meeting societal needs
- Planetary boundaries set limits for human activities, uncertainty does not warrant inaction
- MFA, LCA, indicators used to quantify environmental impact of human activities
- Circular economy is part of the solution, but vulnerable to rebound effects
- ICT has broad, complex impacts on society

Exam(ple) questions

- 1. In the lifecycle analysis of an electronic device (e.g. a phone), which phase usually has the highest environmental footprint?**
 - a. Production
 - b. Transport
 - c. Use
 - d. End of life
- 2. Methane has a GWP of 20. A certain data center consumes 0.3kWh per unit of compute, and is powered by a power plant that emits 1kg of methane and 10kg of CO₂ per kWh produced. What is the carbon intensity of the data center in kgCO₂-eq/compute?**
 - a. 3kgCO₂-eq/compute
 - b. 6kgCO₂-eq/compute
 - c. 9kgCO₂-eq/compute
 - d. 20kgCO₂-eq/compute

Exam(ple) questions

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 - ~~b. 6kgCO₂-eq/compute~~
 - c. 9kgCO₂-eq/compute
 - ~~d. 20kgCO₂-eq/compute~~

Feedback

- We need your feedback to continue improving the course!
- Feedback on Lecture 2 slides/notes:



Announcements

- Project groups formed, one-person groups will be consolidated
- Tomorrow, 15:45-17:30: assignment 'office hours', we are available online for problems, ideas, or feedback
- Midterm presentation (mandatory, ungraded): formative assessment, expect feedback
 - Thursday, October 8, 15:45, EEMCS Hall F
 - Slides: introduction, chosen topic, group progress
- Final presentation (mandatory, graded):
 - Friday, October 22, 15:45-17:45, EEMCS Hall F
 - Format to follow

Thank you for your attention!

Next lecture: Sustainable Software

Thursday, September 17, 15:45-17:30, EEMCS Hall F

- Sustainable software: concept, origins, landscape
- Experimental assessment of software energy efficiency
- Software development practices, impact on sustainability
- Energy efficiency differences across programming languages

Questions?